



Mathematics 2 (5EZB0) Lecture 13

Koondi Mitra

Mathematics & Computer Science Department

Surface integral

- Let $\mathbf{r}(u, v) = x(u, v) \hat{\mathbf{i}} + y(u, v) \hat{\mathbf{j}} + z(u, v) \hat{\mathbf{k}}$ describe a parametric surface $\mathcal{S}$ for $(u, v) \in \mathcal{D}$. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function.
- The surface integral of f is then

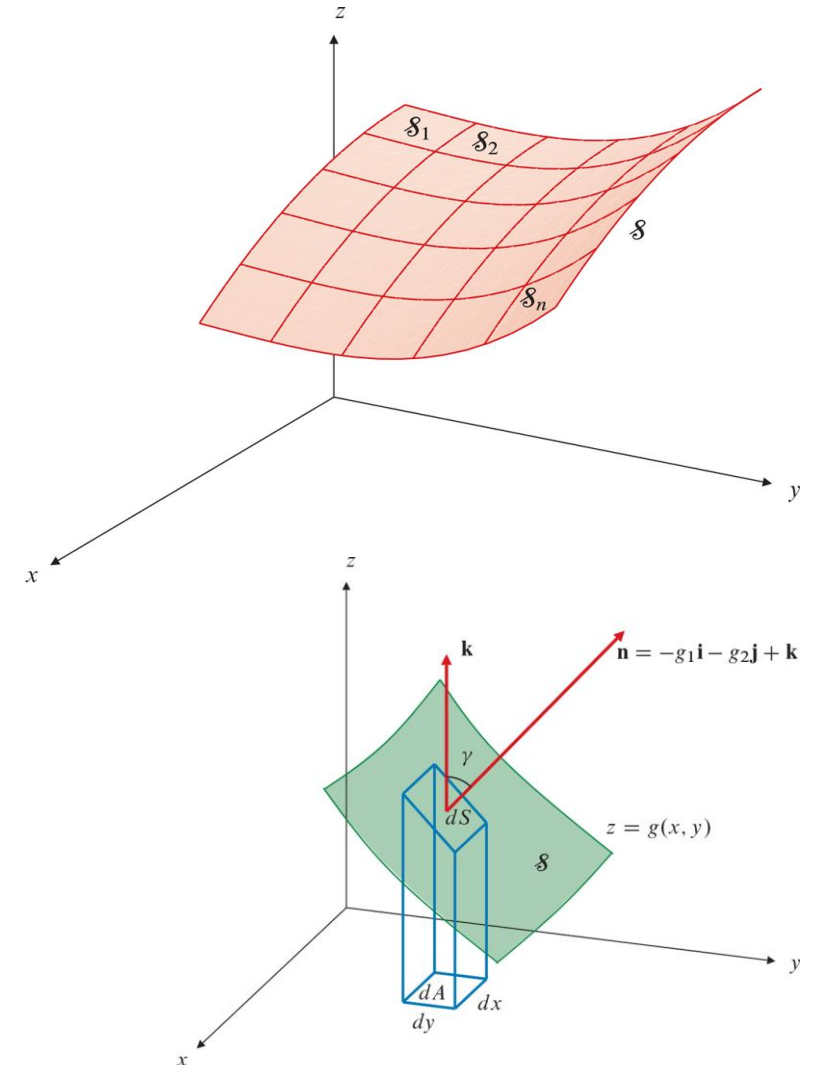
$$\iint_{\mathcal{S}} f(x, y, z) dS = \iint_{\mathcal{D}} f(\mathbf{r}(u, v)) \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| du dv$$

$$\text{where } \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} = \det \begin{pmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{pmatrix} = \frac{\partial(y, z)}{\partial(u, v)} \hat{\mathbf{i}} + \frac{\partial(z, x)}{\partial(u, v)} \hat{\mathbf{j}} + \frac{\partial(x, y)}{\partial(u, v)} \hat{\mathbf{k}}.$$

Remark

- To find the area of $\mathcal{S}$, put $f = 1$
- The formula of area coincides with the formula derived before of surface area of a graph $\mathbf{r}(x, y) = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + f(x, y) \hat{\mathbf{k}}$,

$$|\mathcal{S}| = \iint_{\mathcal{D}} \sqrt{1 + |\nabla f(x, y)|^2} dx dy$$



Oriented surfaces

Oriented surfaces

A smooth surface $\mathcal{S}$ is *orientable* if there exists a unit normal field $\hat{\mathbf{N}}: \mathcal{S} \rightarrow \mathbb{R}^d$ such that for any $\mathbf{P} \in \mathcal{S}$, $\hat{\mathbf{N}}(\mathbf{P})$ varies continuously with $\mathbf{P}$. The field $\hat{\mathbf{N}}$ is called an *orientation*.

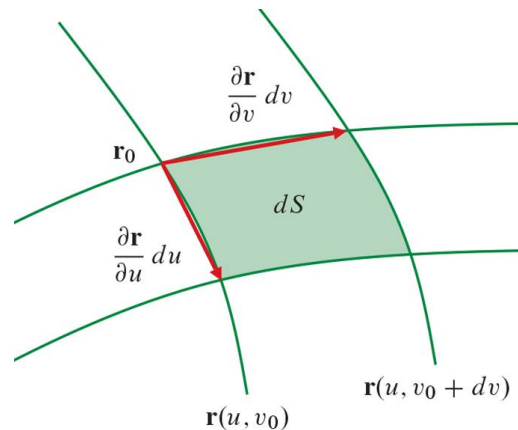
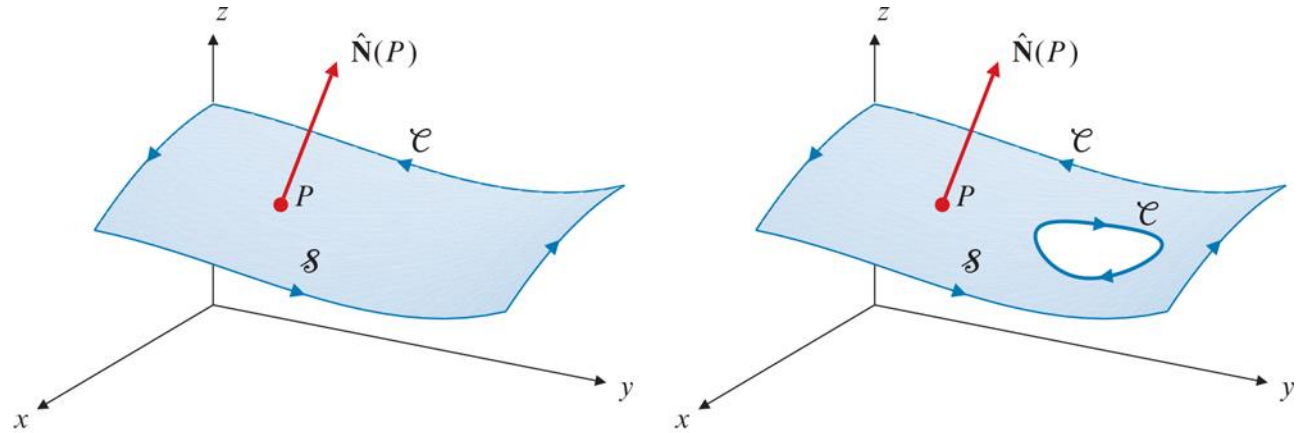
The side which points along $\hat{\mathbf{N}}(\mathbf{P})$ is called the *positive side*, the other side is *negative side*.

➤ For a parametric surface $\mathbf{r}(u, v)$:

$$\hat{\mathbf{N}}(\mathbf{r}(u, v)) = \pm \frac{\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}}{\left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right|}$$

➤ For a graph $z = f(x, y)$

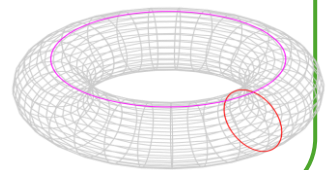
$$\hat{\mathbf{N}}(\mathbf{r}(u, v)) = \pm \frac{-\nabla f + \hat{\mathbf{k}}}{\sqrt{1 + |\nabla f|^2}}$$



Question 1

Are the following surfaces orientable?

1. Torus
2. A cone



Flux integrals

- Let $\mathbf{r}(u, v) = x(u, v) \hat{\mathbf{i}} + y(u, v) \hat{\mathbf{j}} + z(u, v) \hat{\mathbf{k}}$ describe an orientable surface for $(u, v) \in \mathcal{D}$.
- Let $\hat{\mathbf{N}}: \mathcal{S} \rightarrow \mathbb{R}^3$ be the orientation
- Let $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$ be a vector field
- The vector surface area element is defined as

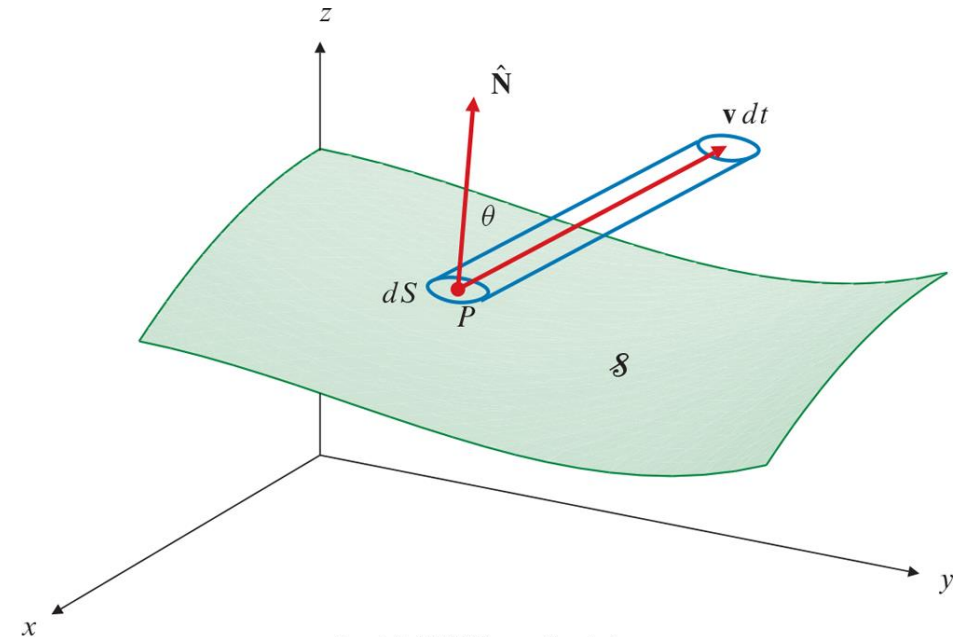
$$d\mathbf{S} = \hat{\mathbf{N}} dS = \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv$$

- Then, the flux integral is

$$\iint_{\mathcal{S}} \mathbf{f} \cdot d\mathbf{S} = \iint_{\mathcal{D}} \mathbf{f}(\mathbf{r}(u, v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv$$

- If $\mathcal{S}$ is a closed surface, then the integral is written as

$$\oiint_{\mathcal{S}} \mathbf{f} \cdot d\mathbf{S}$$



Copyright © 2022 Pearson Canada Inc.

Question 2.1

For the gravitational field

$$\mathbf{F} = -\frac{Cq\mathbf{r}}{|\mathbf{r}|^3}$$

find the flux integral over the surface of the

1. the unit sphere.
2. The cube $[-a, a]^3$

Question 2.2

The flow across the surface of a sieve $z = 1 - x^2 - y^2$ for $z \geq 0$, under the helical vortex

$$\mathbf{f} = -y \hat{\mathbf{i}} + x \hat{\mathbf{j}} + \hat{\mathbf{k}}$$

Gradient, Divergence, and Curl

Gradient

Let $\mathcal{D} \subseteq \mathbb{R}^3$. For $f: \mathcal{D} \rightarrow \mathbb{R}$:

$$\mathbf{grad}(f) = \nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{i}} + \frac{\partial f}{\partial y} \hat{\mathbf{j}} + \frac{\partial f}{\partial z} \hat{\mathbf{k}}$$

Divergence

For $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$:

$$\mathbf{div}(\mathbf{f}) = \nabla \cdot \mathbf{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

Curl

For $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$:

$$\begin{aligned} \mathbf{curl}(\mathbf{f}) &= \mathbf{rot}(\mathbf{f}) = \nabla \times \mathbf{f} \\ &= \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \hat{\mathbf{i}} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \hat{\mathbf{j}} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \hat{\mathbf{k}} \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial_x & \partial_y & \partial_z \\ f_1 & f_2 & f_3 \end{vmatrix} \end{aligned}$$

Maxwell's equations of Electromagnetism

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

Navier-Stokes equations of Fluid mechanics

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p(\rho) + \mu \nabla \cdot \nabla \mathbf{u} + \mathbf{f}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

Physical interpretation

Divergence

For a flow field $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$:

$$\mathbf{div}(\mathbf{f}) = \nabla \cdot \mathbf{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

signifies how much flow is created per unit volume.

The divergence theorem (Not in syllabus)

For a closed surface $\mathcal{S}$ enclosing a domain $\mathcal{D}$:

$$\oint_{\mathcal{S}} \mathbf{f} \cdot d\mathbf{S} = \iiint_{\mathcal{D}} \nabla \cdot \mathbf{f} \, dV$$

Curl

For a flow field $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$:

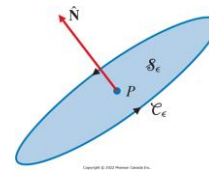
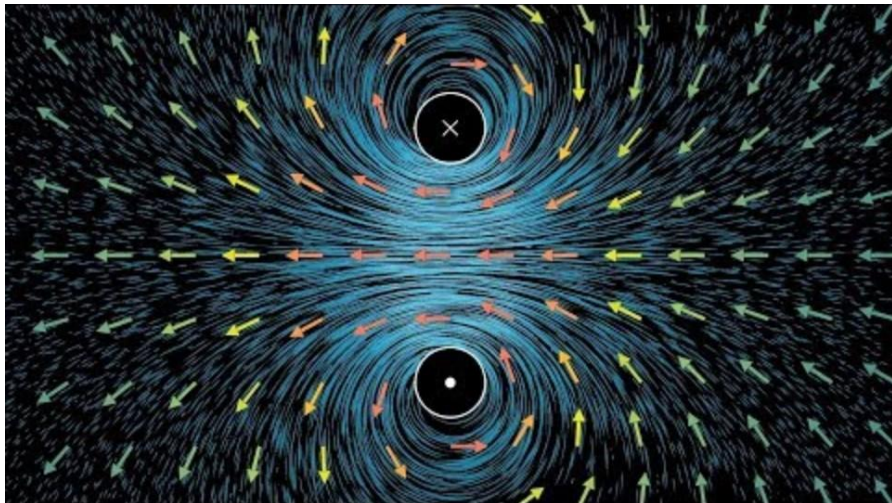
$$\mathbf{curl}(\mathbf{f}) = \mathbf{rot}(\mathbf{f}) = \nabla \times \mathbf{f}$$

represents the rotational density of the field.

Stokes' theorem (Not in syllabus)

For a smooth oriented surface $\mathcal{S}$ with boundary $\mathcal{C}$:

$$\oint_{\mathcal{C}} \mathbf{f} \cdot d\mathbf{r} = \iint_{\mathcal{S}} (\nabla \times \mathbf{f}) \cdot d\mathbf{S}$$



Watch a very nice
video on the
physical
interpretation:

[youtube.com/watch?v=rB83DpBJQsE](https://www.youtube.com/watch?v=rB83DpBJQsE)

Remarks

1. For a rotational field $\mathbf{f} = \boldsymbol{\omega} \times \mathbf{r}$, we have
$$\nabla \times \mathbf{f} = 2\boldsymbol{\omega}$$
2. Curl of conservative field is 0
$$\nabla \times \nabla f = 0$$
3. Divergence of curl is zero:
$$\nabla \cdot (\nabla \times \mathbf{f}) = 0$$

Gradient, Divergence, and Curl

Gradient

Let $\mathcal{D} \subseteq \mathbb{R}^3$. For $f: \mathcal{D} \rightarrow \mathbb{R}$:

$$\mathbf{grad}(f) = \nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{i}} + \frac{\partial f}{\partial y} \hat{\mathbf{j}} + \frac{\partial f}{\partial z} \hat{\mathbf{k}}$$

Divergence

For $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$:

$$\mathbf{div}(\mathbf{f}) = \nabla \cdot \mathbf{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

Curl

For $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$:

$$\mathbf{curl}(\mathbf{f}) = \mathbf{rot}(\mathbf{f}) = \nabla \times \mathbf{f}$$

$$\begin{aligned} &= \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \hat{\mathbf{i}} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \hat{\mathbf{j}} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \hat{\mathbf{k}} \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial_x & \partial_y & \partial_z \\ f_1 & f_2 & f_3 \end{vmatrix} \end{aligned}$$

Question 3.1

Find the div and curl of the flow field

$$\mathbf{F} = -\frac{Cq\mathbf{r}}{|\mathbf{r}|^3}$$

Question 3.2

Study Example 6 in Chapter 17.1. What is the curl of the vector field

$$\mathbf{f} = x \hat{\mathbf{j}}$$

Gradient, Divergence, and Curl

Gradient

Let $\mathcal{D} \subseteq \mathbb{R}^3$. For $f: \mathcal{D} \rightarrow \mathbb{R}$:

$$\mathbf{grad}(f) = \nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{i}} + \frac{\partial f}{\partial y} \hat{\mathbf{j}} + \frac{\partial f}{\partial z} \hat{\mathbf{k}}$$

Divergence

For $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$:

$$\mathbf{div}(\mathbf{f}) = \nabla \cdot \mathbf{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

Curl

For $\mathbf{f}: \mathcal{D} \rightarrow \mathbb{R}^3$:

$$\mathbf{curl}(\mathbf{f}) = \mathbf{rot}(\mathbf{f}) = \nabla \times \mathbf{f}$$

$$\begin{aligned} &= \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \hat{\mathbf{i}} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \hat{\mathbf{j}} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \hat{\mathbf{k}} \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} \end{aligned}$$

More identities

Product rule

- (a) $\nabla(\phi\psi) = \phi \nabla\psi + \psi \nabla\phi$
- (b) $\nabla \cdot (\phi\mathbf{f}) = \nabla\phi \cdot \mathbf{f} + \phi \nabla \cdot \mathbf{f}$
- (c) $\nabla \times (\phi\mathbf{f}) = \nabla\phi \times \mathbf{f} + \phi \nabla \times \mathbf{f}$
- (d) $\nabla \cdot (\mathbf{f} \times \mathbf{g}) = (\nabla \times \mathbf{f}) \cdot \mathbf{g} - \mathbf{f} \cdot (\nabla \times \mathbf{g})$

Second order derivatives

- (a) $\nabla \times \nabla f = 0$
- (b) $\nabla \cdot (\nabla \times \mathbf{f}) = 0$
- (c) $\nabla \times (\nabla \times \mathbf{f}) = \nabla(\nabla \cdot \mathbf{f}) - \nabla^2 \mathbf{f}$

$\nabla^2 f = \nabla \cdot \nabla f$ is called the Laplacian of f

Question 4

Find the div and curl of the vector field

$$\mathbf{f}(r, \theta) = -\sin \theta \hat{\mathbf{i}} + \cos \theta \hat{\mathbf{j}}$$